Ujwal Mahajan

AI / GenAI Engineer · Agentic AI & Multi-Agent Systems · LLM Engineering
2023bec049@sggs.ac.in — +91 8459337231 — LinkedIn — GitHub — Portfolio

Education

B.Tech. Electronics & Telecommunication Engineering *Jan 2023 – Present*
Shri Guru Gobind Singhji Institute of Engineering & Technology, Nanded, Maharashtra

Experience

AI Engineering Intern – BluQ AI *3 Months*
AI-powered construction project management

- Engineered full-stack features (React, Fastify, Supabase); architected WhatsApp/email pipelines automating daily progress capture across 100+ concurrent contractor–PM interactions.
- Designed an iOS/Android application blueprint using React Native/Expo, Supabase Auth, and Backend-for-Frontend (BFF) API strategy.
- Contributed to 5+ architecture decisions with founders covering AI delay prediction, per-user rate-limiting, 300 req/min workload handling, and backend scalability.
- Conducted international market research across 4 countries; sourced and validated 21,000+ firms across 10+ industry directories.

AI / LLM Projects

Pythia-160M – Small Language Model from Scratch *ONGOING* [GitHub]

- Building a ∼160M-parameter Python-focused language model from scratch covering transformer architecture, attention, tokenizer infrastructure, training, and evaluation pipelines.
- Developing an AST-based validation and quality-first code-generation pipeline to prevent syntactically invalid Python from entering downstream processing under constrained local compute.

GPT-2 Architecture – LLM from Scratch [GitHub]

- Built the complete GPT-2 pipeline through Vizuara.ai's LLM-from-scratch course, implementing tokenizer design, embeddings, multi-head attention, GELU, transformer blocks, forward pass, and training.
- Implemented the complete data preparation → tokenizer → embeddings → attention → transformer → training pipeline in PyTorch.

CodeSensei – Multi-Agent Code Intelligence Platform [GitHub] [Live Demo]

- Architected a 5-agent AI orchestration framework (structure, behavior, risk, execution, impact) with structured outputs and synthesizer-based merging; analyzes 5–10K LOC across 50–100 files.
- Built a D3.js code knowledge graph and execution tracing interface using React, TypeScript, Node.js, and Gemini API.

LedgerLens – AI Financial Investigation Platform [GitHub]

- Engineered an end-to-end financial investigation workflow: Reconciliation Engine → Anomaly Detector → Policy Engine → Case Management → Hypothesis Engine → ReAct-style Investigation Agent.
- Stack: FastAPI, SQLite, Streamlit; performs hypothesis-driven agentic reasoning over flagged financial transactions.

TalkToWeb – Agentic Browser Automation [GitHub]

- Converts natural-language commands to executable multi-step browser workflows using LangChain, Gemini, and Playwright for navigation, form filling, scraping, intent parsing, and action planning.
- Built modular skills, headless browser control, voice-interface experimentation, and FastAPI-backed action execution using Python, LangChain, Gemini, Playwright, and FastAPI.

Leadership & Achievements

Lead Organizer – RMAGEDDON 2026 *RNXG Robotics Club*

- Led a 70-member organizing team for a national-level robotics event across 5 competitions; secured ∼2 lakh sponsorships with ∼5,000 attendees.

Finalist – International Rover Challenge 2025

- Contributed to autonomous rover development covering navigation, object detection, sensor integration, and control; presented the rover and technical approach to judges.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++
AI / LLM: PyTorch, Transformers, LangChain, LangGraph, Gemini API, OpenAI API, Hugging Face, Agentic AI, Multi-Agent Systems, RAG, Semantic Search
LLM Engineering: Attention Mechanisms, Tokenization, Embeddings, Model Training, Fine-tuning, Evaluation, Small Language Models
Frontend / Backend: React, React Native/Expo, FastAPI, Fastify, Node.js
Data / Tools: FAISS, ChromaDB, Pinecone, Supabase, SQLite, Neo4j, Docker, Git, Playwright, D3.js, Streamlit

Certifications

Building a Large Language Model from Scratch Vizuara.ai
Prompt Engineering for Developers DeepLearning.AI